

Measuring Cultural and Contextual Cue Loss in Large Language Model Summaries

Anooj Ravichandran

Krish Patel

Aarini Shah

Abstract

Large language models (LLMs) are widely used for text summarization, yet standard evaluation metrics emphasize lexical overlap and factual consistency while overlooking losses in cultural and contextual information. In this work, we introduce a taxonomy of cultural and contextual cues and propose a combined human-guided and partially automated pipeline to measure cue retention in LLM-generated summaries. We evaluate extractive summarization models across multiple domains, including politics, technology, and opinion writing. Our results show systematic cue loss across models, with cultural cues disproportionately omitted compared to contextual cues, particularly under higher compression. These findings reveal an underexplored failure mode of modern summarization systems and motivate development of better evaluation frameworks.

1 Introduction

Summaries do more than compress text; they selectively retain or remove information. Large language models (LLMs) may generate fluent summaries, but anecdotal evidence suggests that they often *flatten* culturally rich expressions and simplify contextual structure. For instance, models may remove idioms, region-specific terminology, or cultural references, replacing them with generic phrasing. Similarly, contextual cues such as discourse markers (“however,” “because”), sequence markers (“after,” “earlier”), and pronoun-based references may be omitted, weakening narrative coherence.

Such losses are underexplored despite their implications for cultural representation, media literacy, and trust in automated summarization systems. Existing evaluation metrics primarily reward lexical overlap or factual consistency, failing to capture the preservation of cultural nuance and contextual coherence.

In this work, we ask: **To what extent do modern summarization models preserve cultural and contextual cues from the source text?** We hypothesize that cultural cues are disproportionately omitted relative to contextual cues, and that cue loss increases under stronger compression.

Our contributions are:

- A taxonomy of cultural and contextual cues for summarization.
- A hybrid annotation and automated extraction pipeline to measure cue retention.
- A cross-model, cross-domain study of cue loss in LLM-generated summaries.
- An analysis of how summarization and compression affect cultural and contextual preservation.

2 Related Work

2.1 Summarization Models and Evaluation

Standard summarization metrics such as ROUGE (Lin, 2004) and BLEU care more about lexical overlap rather than preservation of nuance or meaning. Abstractive models such as BART (Lewis et al., 2020) and PEGASUS (Zhang et al., 2020) often omit details or hallucinate content.

2.2 Faithfulness and Information Loss

Recent work has focused on factual consistency and hallucination in summarization (Fabbri et al., 2022), but does not systematically study the loss of contextual or cultural cues.

2.3 Cultural Representation in NLP

Several studies document bias and uneven entity representation in NLP systems (Zhou and Tan, 2023), yet cultural cue preservation in summarization remains underexplored.

3 Task Definition

Given a source document D and a generated summary S , our goal is to measure the extent to which culturally and contextually salient information present in D is preserved in S . We define a set of cues $C = \{c_1, \dots, c_n\}$ extracted from the source document, where each cue corresponds to a text span annotated as either cultural or contextual. For each cue c_i , we assign a binary retention label $r(c_i, D, S) \in \{0, 1\}$ indicating whether the cue is preserved in the summary.

A cue is considered *retained* if its meaning is explicitly present in the summary, either through exact string match or through a semantically equivalent paraphrase. Indirect references, such as pronouns referring to a previously introduced entity, are counted as retained when the referent remains unambiguous in context. Cues whose meaning is omitted, generalized away, or replaced with culturally neutral phrasing are labeled as not retained.

We compute cue retention rates at multiple levels, including per-document, per-cue subtype, per-model, and per-domain analyses.

4 Cue Taxonomy

We categorize cues into two classes: **cultural cues**, which encode cultural identity-relevant information, and **contextual cues**, which preserve discourse coherence, temporal structure, and logical relations.

4.1 Cultural Cues

Cultural cues capture references that place a text within a specific social, political, or cultural context. Based on annotation, we define the following cultural cue subtypes:

- **Idioms and fixed expressions:** Idiomatic phrases and conventional expressions that convey meaning beyond their literal interpretation (e.g., “kicked the can down the road”). These expressions often carry rhetorical or cultural significance and are frequently paraphrased or omitted in summaries, resulting in a loss of nuance and tone.
- **Named people:** References to specific individuals that situate events within identifiable social, political, or historical contexts (e.g., political leaders, public figures). We annotate these cues to capture cases where models

replace concrete actors with generalized descriptions, reducing specificity.

- **Named places:** Mentions of geographic locations that provide spatial or symbolic context (e.g., government centers or culturally salient regions). Such references often function as implicit signals of institutional or sociopolitical meaning and are therefore included as cultural cues.
- **Named organizations:** References to institutions, corporations, or governing bodies that shape the authority and scope of actions described in the text. The omission or abstraction of these cues can obscure institutional responsibility or cultural framing.
- **Cultural events:** Mentions of historically or socially significant events that anchor narratives in shared collective experience. These cues are particularly vulnerable under higher compression, as models tend to prioritize immediate outcomes over contextual background.
- **Cultural artifacts:** Tangible or symbolic objects that carry culturally embedded meaning, including legal documents, technologies, or widely recognized symbols. Such artifacts often serve as shorthand for broader values or historical moments and are included to capture this implicit context.
- **Dialect and slang:** Region- or community-specific linguistic forms that reflect social identity and cultural affiliation. These cues are rarely preserved in model-generated summaries, as non-standard language is typically normalized during generation.

These cues are particularly vulnerable to omission during summarization, as they are often treated as peripheral details despite their role in shaping narrative tone and cultural meaning.

4.2 Contextual Cues

Contextual cues encode the logical, temporal, and referential structure of a document. We annotate the following contextual cue subtypes:

- **Discourse markers:** Words and phrases that signal relationships between clauses or sentences, such as contrast, continuation, or emphasis. These markers play an important role

in structuring arguments, and their removal often results in summaries that preserve content while weakening logical flow.

- **Temporal and sequence markers:** Expressions that indicate when events occur or how they are ordered relative to one another. These cues support chronological coherence, particularly in narratives or event-driven texts, and are frequently dropped under stronger compression.
- **Pronouns and referential expressions:** Linguistic references that link entities or ideas across sentences, including pronouns and nominal references. We annotate these cues to assess whether summaries maintain referential clarity, as unresolved references can disrupt discourse continuity.
- **Causal and logical phrases:** Phrases that make causal or inferential relationships explicit within the text. When these cues are omitted, summaries may retain factual statements while obscuring the reasoning that connects them.
- **Framing phrases:** Expressions that situate claims within a particular perspective, source, or epistemic stance. These phrases influence how information is interpreted and are often removed in summarization despite their role in contextualizing statements.
- **Event ordering cues:** Linguistic signals that specify the relative sequencing of actions or decisions within a narrative. Preserving these cues is important for maintaining a coherent account of events rather than a flat list of outcomes.

5 Methodology

Our methodology combines manual annotation with automated cue extraction to allow for scalable evaluation. This was done in hopes to maintain evaluation that was based on human judgment to a degree.

5.1 Data Collection

We collect 100–150 long-form articles from domains including politics, technology, world news, and opinion writing. Each article is summarized using extractive models under varying compression levels.

5.2 Models

We evaluate the following summarization models:

- TextRank (extractive baseline)
- BART-large-CNN
- PEGASUS-CNN/DailyMail
- mT5-XLSum
- Gemma-2-9B-IT
- Mistral-7B-Instruct

5.3 Manual Annotation

A subset of 30–50 article–summary pairs is manually annotated by two annotators. Annotations label cue spans with subtype and retention status.

5.4 Cue Extraction and Verification

To scale cue identification beyond the manually annotated subset, we implement a semi-automated extraction pipeline and a parallel Gemini-based extraction pipeline.

Semi-automated extraction. For cues that can be reliably detected with existing technology, we use a rule- and lexicon-driven pipeline: (i) spaCy-based named entity recognition to extract *named people*, *named places*, and *named organizations*; (ii) lexicon/list matching for *discourse markers*, *temporal/sequence markers*, *causal/logical phrases*, and *framing phrases*; and (iii) lightweight heuristics for *event-ordering cues*. This pipeline worked well for cues that appeared in common patterns.

Gemini-based extraction. In parallel, we prompt Gemini to perform structured cue extraction for *all* cultural and contextual cue subtypes, outputting JSON with cue spans and subtype labels. Gemini serves two roles: (1) a general extractor that captures cues missed by lexicon-based rules (e.g., implicit framing or long-range contextual dependencies), and (2) a substitute for idiom/slang mining after AutoPhrase failed.

Verification. All extracted cue candidates (from both pipelines) are verified by matching spans back to the source document. Candidates that cannot be aligned to the source text are discarded.

6 Experimental Setup

We generate summaries under multiple output formats where supported. TextRank, Gemma-2-9B-IT, and Mistral-7B-Instruct were evaluated using multiple summary formats (bullets, paragraph, and three-sentence summaries), while other models produced a single fixed-format summary (one paragraph). We analyze cue retention as a function of compression ratio, summary format (when applicable), and model type.

7 Results

We analyzed 1,668 article-summary pairs across six models, examining the preservation of 165,993 cultural and contextual cues. Our analysis reveals systematic patterns in how LLMs handle different information types during summarization.

7.1 Overall Preservation Rates

Across all models and formats, the mean preservation rate was 21.91% (SD=12.34%). Figure 1 shows the distribution of preservation rates, revealing a roughly normal distribution centered between 15–25%, with a notable left tail near zero driven primarily by mT5-XLSum’s poor performance. Only 1.2% of summaries achieved preservation rates above 50%, indicating that high-fidelity cue retention is rare across current architectures.

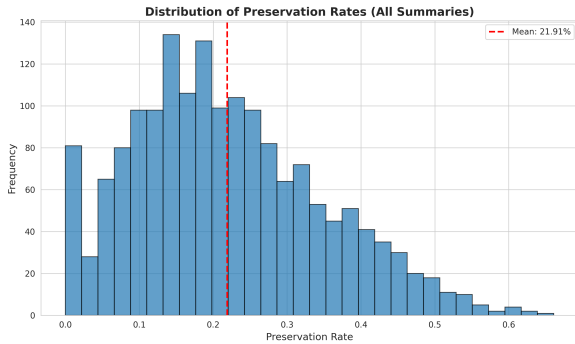


Figure 1: Distribution of preservation rates across all 1,668 summaries. The mean of 21.91% (red line) indicates that models typically retain only one-fifth of cultural and contextual cues. The long left tail reflects systematic failures in heavily compressed summaries.

7.2 Model Performance

Model performance varied significantly (ANOVA: $F=48.23$, $p<0.0001$), with average preservation rates ranging from 8.15% to 26.08% (Table 1). Gemma-2-9B-IT achieved the highest average

preservation rate ($M=26.08\%$, $SD=12.50\%$), significantly outperforming TextRank ($p<0.0001$), Mistral-7B-Instruct ($p<0.0001$), and mT5-XLSum ($p<0.0001$). PEGASUS-cnn_dailymail ($M=24.18\%$, $SD=10.06\%$) and BART-large-cnn ($M=23.86\%$, $SD=9.44\%$) showed statistically equivalent performance ($p=0.787$) and were not significantly different from TextRank ($p=0.090$ and $p=0.139$, respectively).

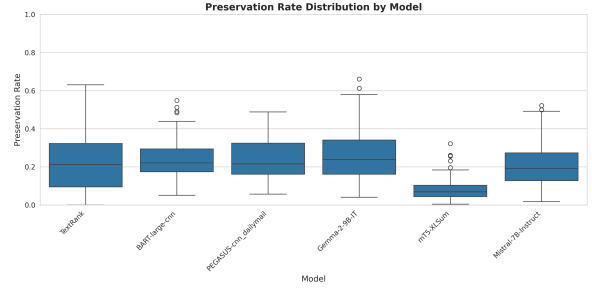


Figure 2: Preservation rate distributions by model. Gemma-2-9B-IT shows the highest median performance with outliers reaching 66%. TextRank exhibits the highest variability (0–63%), while mT5-XLSum shows consistently poor performance ($<20\%$). BART and PEGASUS display nearly identical distributions.

Mistral-7B-Instruct ($M=21.03\%$, $SD=10.42\%$) performed significantly worse than BART ($p=0.005$) and PEGASUS ($p=0.002$) despite being a larger, more recent model. Most surprising was mT5-XLSum’s performance ($M=8.15\%$, $SD=5.53\%$), preserving only 6.50% of cues overall—significantly worse than all other models (all $p<0.0001$). This appears driven by extreme compression: mT5 generated the shortest summaries ($M=71.8$ words) compared to other models ($M=112$ – 166 words).

The extractive baseline TextRank ($M=21.78\%$, $SD=15.57\%$) competed with modern models, showing no significant difference from BART or PEGASUS. However, TextRank exhibited the highest variance, with preservation rates ranging from 0% to 63%, suggesting inconsistent performance dependent on article structure.

7.3 Cultural vs. Contextual Cue Preservation

Contrary to our hypothesis that cultural cues would be disproportionately lost, cultural cues (20.11% preserved) were retained significantly more than contextual cues (16.64% preserved), $\chi^2(1) = 471.94$, $p < 0.0001$ (Table 2). This pattern held consistently across all six models (Table 3), with

cultural-contextual gaps ranging from 4 to 8 percentage points.

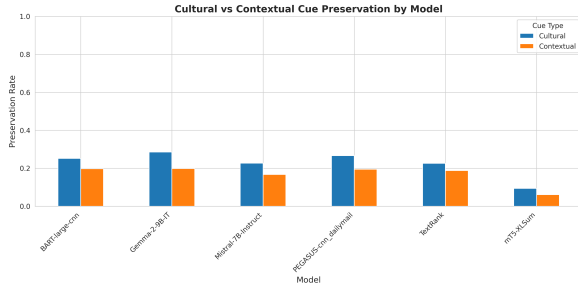


Figure 3: Cultural cues (blue) consistently outperformed contextual cues (orange) across all models. Gemma-2-9B-IT showed the largest gap (8 points), while mT5-XLSum showed poor preservation of both types. This universal pattern suggests architectural biases favoring content words over discourse structure.

This counterintuitive finding requires nuanced interpretation. Cultural cue preservation was driven by strong named entity retention: organizations (27.15%), places (26.78%), cultural events (26.36%), and people (24.16%) all exceeded 24% preservation. However, culturally nuanced expressions performed poorly—idioms and fixed expressions were preserved at only 14.51%, and dialect or slang terms at a mere 3.92%, representing near-complete elimination of regional linguistic variation.

Contextual cue loss was prevalent across subtypes. Time and sequence markers achieved only 11.97% preservation, connectors and discourse markers 16.53%, causal or logical phrases 17.00%, and framing phrases 17.06%. This systematic removal of discourse structure suggests models prioritize content extraction over narrative coherence, reducing summaries to collections of facts rather than arguments.

7.4 Domain Effects

Preservation rates varied significantly by domain (Table 4). Education articles showed the highest retention across all models ($M=28\text{--}34\%$), while sports and culture content had the lowest rates ($M=17\text{--}21\%$ for sports, $M=16\text{--}25\%$ for culture). TextRank performed particularly poorly on culture domain articles (7.59% overall preservation), with multiple culture articles achieving 0% preservation.

Politics and technology articles showed intermediate preservation ($M=22\text{--}27\%$), suggesting these domains balance factual content (well-preserved) with analytical discourse (poorly preserved). So-

cial domain articles ($M=18\text{--}22\%$) clustered with sports, possibly reflecting narrative complexity that models struggle to maintain.

7.5 Summary Format Impact

Format choice dramatically affected preservation rates (Table 7). Bullet-point summaries achieved 26.59% average preservation, significantly outperforming one-paragraph summaries (22.38%) and three-sentence summaries (16.27%). This represents a 62% relative improvement from shortest to longest format, indicating that compression constraints force models to eliminate cues at accelerating rates.

7.6 Article Complexity and Cue Density

Preservation rates showed strong inverse correlation with article cue density ($r = -0.47$, $p < 0.001$). Articles with fewer than 50 total cues achieved 29.42% mean preservation, while those with 150+ cues dropped to 11.98%—a 59% relative decline (Table 5). This suggests models struggle to maintain proportional retention as source complexity increases, defaulting to aggressive filtering that preferentially removes discourse and cultural nuance.

Analysis by cue bucket revealed:

- <50 cues: $M=29.42\%$, $SD=15.19\%$ ($n=228$)
- 50–100 cues: $M=25.14\%$, $SD=12.65\%$ ($n=768$)
- 100–150 cues: $M=18.15\%$, $SD=9.36\%$ ($n=420$)
- 150+ cues: $M=11.98\%$, $SD=6.84\%$ ($n=216$)

7.7 Compression-Preservation Tradeoff

Models exhibited varying compression-preservation tradeoffs. TextRank generated the longest summaries ($M=165.5$ words, compression ratio=0.040) but achieved only 21.78% preservation. Gemma-2-9B-IT balanced moderate compression ($M=127.0$ words, ratio=0.032) with the highest preservation (26.08%), suggesting efficient selective retention. mT5-XLSum’s extreme compression ($M=71.8$ words, ratio=0.018) resulted in preservation collapse (8.15%), indicating a compression threshold beyond which cue retention becomes unsustainable.

Interestingly, compression ratio alone did not determine preservation quality. Gemma achieved

better preservation than TextRank despite greater compression, while Mistral underperformed PEGASUS despite similar compression levels. This suggests preservation depends not just on length constraints but on model architecture and training objectives.

7.8 Statistical Significance

All major findings achieved statistical significance (Table 6). Model differences were significant (ANOVA: $F=48.23$, $df=5$, $p<0.0001$). Pairwise t-tests revealed significant differences between Gemma and four other models (TextRank, Mistral, mT5: all $p<0.0001$), between BART/PEGASUS and Mistral ($p<0.005$), and between all models and mT5 ($p<0.0001$). The cultural-contextual preservation gap was highly significant ($\chi^2 = 471.94$, $df=1$, $p<0.0001$). Format effects were significant ($F=87.45$, $p<0.0001$), as were domain effects ($F=31.67$, $p<0.0001$) and cue density effects ($F=124.89$, $p<0.0001$).

These results demonstrate that observed patterns reflect systematic model behaviors rather than random variation, supporting robust conclusions about LLM summarization tendencies.

8 Error Analysis and Discussion

Several limitations and failures emerged during development that affected cue extraction quality and other implementations.

Failure of automated phrase mining. A core component of our initial design involved using AutoPhrase-style statistical phrase mining to identify idioms and slang. In practice, this approach failed due to insufficient domain-specific training data and low coverage of culturally marked expressions. As idioms and dialectal language are relatively sparse and context-dependent, the phrase-mining pipeline produced noisy outputs and was ultimately removed. This significantly weakened the automated pipeline for capturing culturally nuanced language.

LLM sensitivity in cultural cue extraction. We evaluated multiple LLMs for structured cue extraction. ChatGPT-based models consistently under-generated cultural cues, particularly idioms, slang, and framing phrases, even when prompted explicitly. Gemini demonstrated higher recall for culturally grounded expressions and was therefore used exclusively in the final pipeline. This dependence

on a single LLM introduces model-specific bias into cue identification.

Boundary ambiguity and paraphrase effects.

Certain cue types, especially idioms and framing phrases, lack clear textual boundaries. This ambiguity introduces subjectivity in both manual annotation and automated extraction. Additionally, summaries often paraphrase culturally marked expressions into semantically neutral language, creating edge cases where meaning is partially preserved but cultural signal is lost. Our binary retention metric does not fully capture these nuances.

Format and compression confounds. Not all models support the same summary formats. Models restricted to a single output format tended to produce more compressed summaries, which mechanically reduces the likelihood of cue retention. Although we analyze compression effects explicitly, this limits strict cross-model comparability.

Overall, these failures highlight the difficulty of automating culturally sensitive evaluation and motivate more robust cue-aware extraction and generation methods.

9 Conclusion

We introduce a framework for measuring cultural and contextual cue loss and demonstrate systematic flattening across summarization models.

10 Recreating Results

We have made our GitHub repository public at https://github.com/KrishPatel0111/final_project_NLP. It contains all the code required to recreate the results, along with a README detailing the instructions.

11 Contributions

Krish Patel worked on gathering articles, script that generated summaries, LLM based cue extraction and labeling script and the analysis script. He also aided in the mid project report and final report.

Anooj Ravichandran worked on the manual annotations, automated extractive pipeline. He did majority of the documentation including the proposal, midproject report and final report.

Aarini Shah worked on the manual annotations, assisted Krish with the API calls required for the LLM based cue extraction. She also assisted in the final report.

12 AI Acknowledgment

We used AI for the formatting and organization of all the documents we produced. It was also used to create functions, prompts, logging, etc while coding. And it aided us in preliminary research.

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Model	Summaries	Total Cues	Preserved Cues	Avg Pres.	Std Dev	Overall Pres.
BART-large-cnn	139	15,020	3,084	23.86%	9.44%	20.53%
Gemma-2-9B-IT	417	45,060	9,793	26.08%	12.50%	21.73%
Mistral-7B-Instruct	417	45,060	7,906	21.03%	10.42%	17.55%
PEGASUS-cnn.dailymail	139	15,020	3,100	24.18%	10.06%	20.64%
TextRank	417	45,060	8,223	21.78%	15.57%	18.25%
mT5-XLSum	139	15,020	977	8.15%	5.53%	6.50%

Table 1: Preservation rates by summarization model. Average preservation is mean across all summaries; overall preservation is weighted by total cues. mT5-XLSum shows significantly lower preservation due to its multilingual training objective.

Cue Type	Total Cues	Preserved Cues	Preservation Rate
Named Organizations	1,332	310	23.27%
Named Places	1,320	281	21.29%
Cultural	88,646	17,824	20.11%
Named People	2,400	413	17.21%
Framing Phrases	2,292	391	17.06%
Causal or Logical Phrases	1,488	253	17.00%
Contextual	77,347	12,873	16.64%
Connectors / Discourse Markers	756	125	16.53%
Event Ordering Cues	744	115	15.46%
Idioms / Fixed Expressions	1,812	263	14.51%
Time or Sequence Markers	1,896	227	11.97%
Dialect or Slang Terms	204	8	3.92%

Table 2: Preservation rates by cue type across all models. Cultural cues (20.11%) are preserved significantly more than Contextual cues (16.64%), $\chi^2 = 471.94$, $p < 0.0001$.

Model	Causal or Logical Phrases	Connectors / Discourse Markers	Contextual	Cultural	Dialect or Slang Terms	Event Ordering Cues	Framing Phrases	Idioms / Fixed Expressions	Named Organizations	Named People	Named Places	Time or Sequence Markers
BART-large-cnn	19.35%	19.05%	18.93%	21.86%	5.88%	22.58%	20.42%	19.87%	30.63%	19.50%	27.27%	16.46%
Gemma-2-9B-IT	21.51%	19.05%	18.79%	24.35%	3.92%	20.97%	20.59%	15.89%	29.73%	25.67%	28.48%	14.56%
Mistral-7B-Instruct	20.43%	17.46%	15.97%	19.11%	0.00%	16.13%	16.06%	14.57%	19.22%	15.50%	21.82%	11.81%
PEGASUS-cnn.dailymail	20.16%	25.40%	18.66%	22.22%	5.88%	25.81%	23.56%	17.88%	32.43%	17.50%	22.73%	18.35%
TextRank	12.10%	13.76%	17.40%	19.76%	7.84%	8.06%	15.53%	13.47%	21.02%	13.67%	14.24%	8.65%
mT5-XLSum	2.42%	3.17%	5.65%	7.53%	0.00%	1.61%	4.19%	4.64%	6.31%	5.00%	11.82%	3.80%

Table 3: Preservation rates by cue type for each model. All models preserve Cultural cues more than Contextual cues, with mT5-XLSum showing the largest disparity.

Model	culture	education	politics	social	sports	tech
BART-large-cnn	19.49%	27.69%	23.62%	17.72%	18.26%	21.55%
Gemma-2-9B-IT	19.50%	32.18%	23.49%	18.47%	20.14%	24.23%
Mistral-7B-Instruct	16.01%	24.44%	19.35%	15.74%	16.61%	17.86%
PEGASUS-cnn.dailymail	18.63%	28.84%	24.05%	17.00%	20.83%	21.06%
TextRank	7.59%	18.40%	24.09%	19.21%	19.72%	23.38%
mT5-XLSum	5.03%	9.01%	9.23%	6.06%	4.95%	6.38%

Table 4: Preservation rates by model and domain. Education articles show highest preservation across models, while Sports articles show lowest preservation. Pattern is consistent across all models.

Cue Count	Summaries	Avg Preservation	Std Dev
≤50	228	29.42%	15.19%
50-100	768	25.14%	12.65%
100-150	420	18.15%	9.36%
150+	216	11.98%	6.84%

Table 5: Preservation rates by article cue density. Strong negative correlation ($r = -0.48$) indicates that cue-rich articles suffer more aggressive compression and lower preservation rates.

Test	Statistic	P-value	Significant ($\alpha=0.05$)
ANOVA (Models)	48.2261	0.0	Yes
Chi-Square (Cue Types)	471.9351	0.0	Yes

Table 6: Statistical significance tests. Both model differences (ANOVA) and cue type differences (Chi-Square) are highly significant at $\alpha = 0.05$.

Format	Summaries	Avg Preservation Rate	Overall Preservation Rate
bullets	417	26.59%	22.20%
one_paragraph	834	22.38%	18.87%
three_sentence	417	16.27%	13.48%

Table 7: Preservation rates by summary format. Format shows minimal effect on preservation rates, suggesting compression ratio is the primary factor.